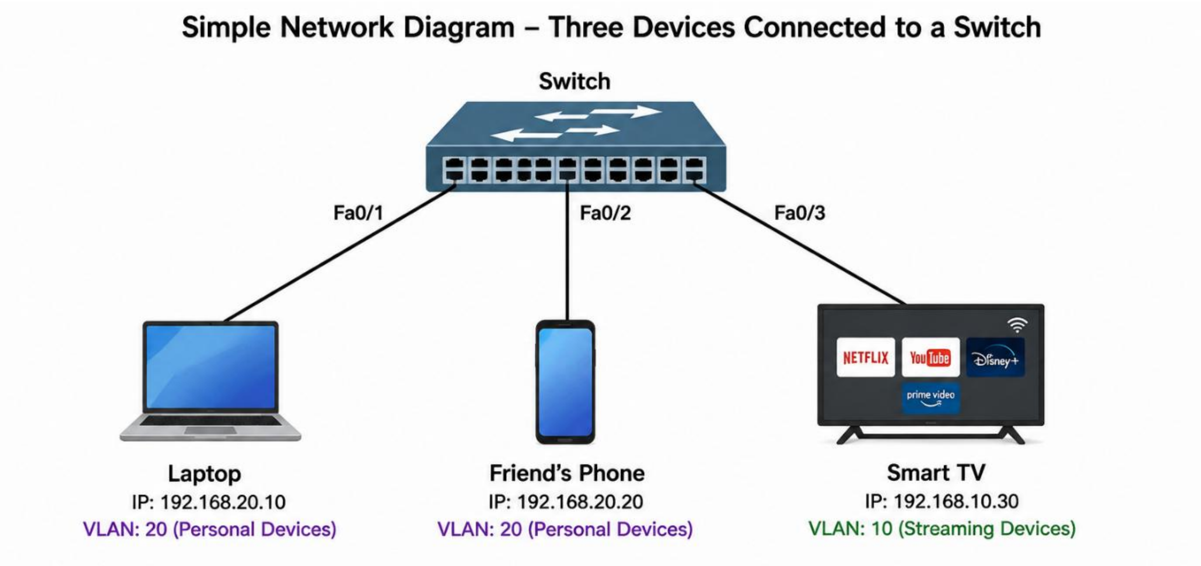
## Switching and VLANS

1) Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).




| **Device** | **IP Address** | **VLAN** | **Purpose** |
|---|---|---|---|
| Laptop | 192.168.20.10 | VLAN 20 | Personal Device |
| Friend's Phone | 192.168.20.20 | VLAN 20 | Personal Device |
| Smart TV | 192.168.10.30 | VLAN 10 | Streaming Device |

2) Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.<br><br><em><strong>Hint:</strong> Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.</em>

```
Switch> enable
Switch# configure terminal

Switch(config)# vlan 10
Switch(config-vlan)# name staff
Switch(config-vlan)# exit

Switch(config)# vlan 20
Switch(config-vlan)# name student
Switch(config-vlan)# exit

Switch(config)# interface range fa0/2-4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit

Switch(config)# interface range fa0/5-6
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
```

```
C:\>ping 10.0.0.51

Pinging 10.0.0.51 with 32 bytes of data:

Reply from 10.0.0.51: bytes=32 time<1ms TTL=128
Reply from 10.0.0.51: bytes=32 time=5ms TTL=128
Reply from 10.0.0.51: bytes=32 time=2ms TTL=128
Reply from 10.0.0.51: bytes=32 time=4ms TTL=128

Ping statistics for 10.0.0.51:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 2ms

C:\>ping 10.0.0.101

Pinging 10.0.0.101 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.0.101:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

```
C:\>ping 10.0.0.102

Pinging 10.0.0.102 with 32 bytes of data:

Reply from 10.0.0.102: bytes=32 time=1ms TTL=128
Reply from 10.0.0.102: bytes=32 time<1ms TTL=128
Reply from 10.0.0.102: bytes=32 time<1ms TTL=128
Reply from 10.0.0.102: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.0.0.51

Pinging 10.0.0.51 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.0.51:
    Packets: Sent = 3, Received = 0, Lost = 3 (100% loss),
```

3) In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

Switch# configure terminal

Switch(config)# interface fa0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# end

Output:
VLAN 10
Fa0/1
Fa0/2

VLAN 20
Fa0/3
Fa0/4
Fa0/5

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.100

Pinging 10.0.0.100 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.0.100:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.0.0.51

Pinging 10.0.0.51 with 32 bytes of data:

Reply from 10.0.0.51: bytes=32 time<1ms TTL=128
Reply from 10.0.0.51: bytes=32 time<1ms TTL=128
Reply from 10.0.0.51: bytes=32 time<1ms TTL=128
Reply from 10.0.0.51: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.51:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.50

Pinging 10.0.0.50 with 32 bytes of data:

Reply from 10.0.0.50: bytes=32 time<1ms TTL=128
Reply from 10.0.0.50: bytes=32 time<1ms TTL=128
Reply from 10.0.0.50: bytes=32 time<1ms TTL=128
Reply from 10.0.0.50: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

4) Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).<br><br><em><strong>Constraint:</strong> Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.</em>

```
Switch> enable
Switch# configure terminal

Switch(config)# vlan 30
Switch(config-vlan)# name SALES
Switch(config-vlan)# exit

Switch(config)# interface range fa0/1-10
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
Switch(config-if-range)# exit

Switch(config)# end
Switch# write memory
```